

INSPECTA Overview

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Introduction

In this document we give an overview of the ‘Integrated Spectral line Processing Engine in C for Telescope data Analysis’, ‘INSPECTA’, software package that can be used to process high frequency-resolution observations recorded in ‘Spectral-Domain Hierarchical Data Format’ (‘SDHDF’) from radio telescopes.

Full details of the SDHDF definition are described in Toomey et. al 2024 (<https://doi.org/10.1016/j.ascom.2024.100804>).

The C code along with a python implementation ‘pyINSPECTA’ (under development) can be found in the ‘sdhdf_tools’ repository: https://bitbucket.csiro.au/projects/CPDA/repos/sdhdf_tools/browse.

Summary of packages in INSPECTA

Identifying/inspecting SDHDF files	
sdhdf_describe	Show the primary metadata for a list of files (including project IDs, source names, observer, etc.)
sdhdf_quickdump	Produce text files containing information within an SDHDF file
sdhdf_identify	Identify a list of files with specified parameters (e.g., source names, project codes)
sdhdf_checkFile	Look for known issues with a file
Plotting/visualising SDHDF files	
sdhdf_plotSpectrum	Interactive visualisation of the spectra for a single band in a single SDHDF file.
sdhdf_plotMultiSpec	Non-interactive plotting of spectra for a single band in one or more data files.
sdhdf_plotWide	Interactive visualisation of the entire band (multiple sub-bands).

sdhdf_plotScan	Plot of the signal strength as a function of time-dependent parameters including pointing directions.
sdhdf_waterfall	Make waterfall-style plots
Manipulating SDHDF files	
sdhdf_extractBand	Extract spectral bands from an SDHDF file to produce a new (smaller) file (e.g., specific zoom bands)
sdhdf_extractDump	Extract spectral time dumps from an SDHDF file to produce a new (smaller) file
sdhdf_modify	Produce a new SDHDF file by modifying the contents of an existing file (i.e., averaging in time or frequency)
sdhdf_sum	Produce a new SDHDF file by summing the spectral dumps in a set of existing files.
sdhdf_join	Join two or more sdhdf files together
Flagging	
sdhdf_flag	Interactive flagging of a data file
sdhdf_autoFlag	Automatic flagging of a data file (including copying of flagged data from a different file)
Conversion tools	
sdhdf_convertFromFITS	Convert to SDHDF-format from other formats
sdhdf_convertFromPsr	
sdhdf_convertTo	Convert from SDHDF-format to other formats (currently just SDFITS)
Processing tools	
sdhdf_onoff	Determine (ON-OFF)/OFF spectra from an ON-source file and an OFF-source file
sdhdf_map	Produce a sky map from multiple input scans
sdhdf_pointing	Produce pointing solutions using multiple scans
sdhdf_gainCurve	
sdhdf_statistics	Determine statistics for specific spectra

Calibration	
sdhdf_calibrate	Calibrates (flux density and polarisation) sdhdf files
sdhdf_plotNoiseCal	Plots and carries out basic processing tasks on the noise source calibration signal
Miscellaneous	
sdhdf_listLines	List the rest frequencies of spectral lines in a given frequency range.

Package descriptions

The packages available as part of the INSPECTA software package are described in alphabetic ordering below.

The example data files used here can be obtained using:

```
$ wget https://data.atnf.csiro.au/download/2021-08-17-16-57-39-320-LvAUALCbnmiQNubljRsa/uwl\_210722\_012146.hdf
```

sdhdf_autoFlag

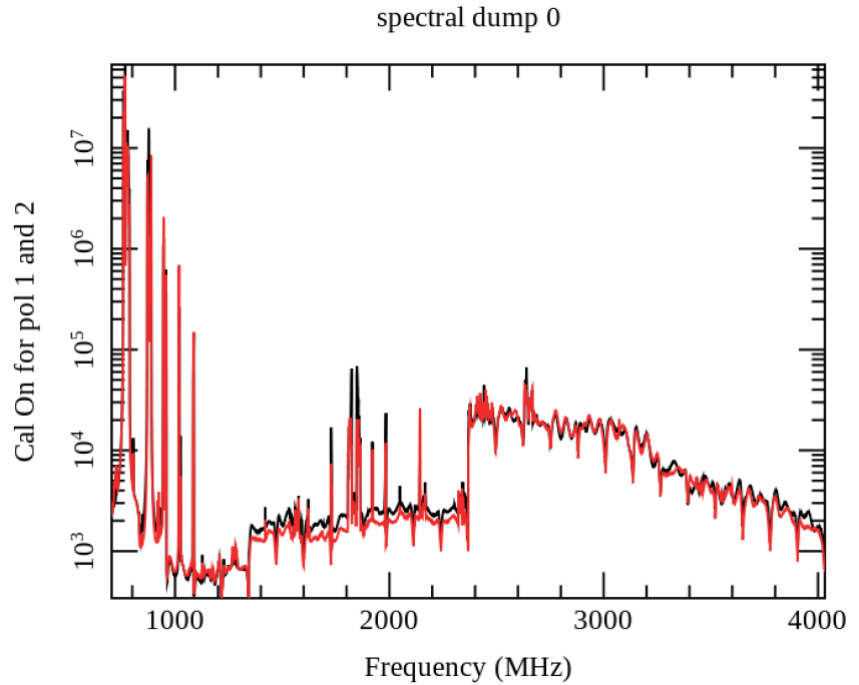
Purpose	Automatically sets the weights to zero for spectral channels known to be affected by radio frequency interference of digital artefacts
File input	Multiple input files can be listed on the command line
Command line arguments	-e <ext> Produces output files with specified extension (default is .autoFlag) -edge <val> Removes “val” percent of each sub-band boundary -from <file> Copies flagging information from that contained in the specified file -h Help information

	-noflags Do not produce a flag table -noweights Do not produce a weights table (note that it doesn't remove a weights table that is already present) -persistent Removes known persistent RFI. The list of RFI is stored in \$SDHDF_RUNTIME/observatory/<id>/<rfi>/persistent.dat -transient Removes transient RFI if it is present. The RFI is identified using the logic described in \$SDHDF_RUNTIME/observatory/<id>/<rfi>/transient.dat
Output	Output files (by default with file extension .autoFlag) for each input file. The weights table will be set to zero for channels that should be flagged. The text output on the screen describes which RFI has been flagged.
Interactive key presses	None
Example	<code>sdhdf_autoFlag -transient -persistent uwl_210722_012146.hdf</code>
Known bugs	

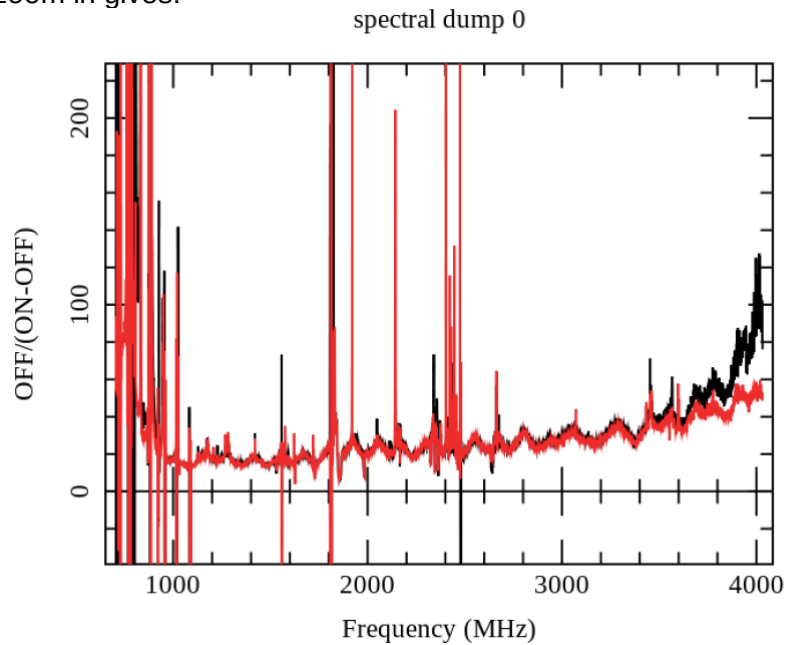
sdhdf_plotNoiseCal

Purpose	Make graphical and interactive plots relating to the noise source
File input	Use -f command line argument for a single input file
Command line arguments	-autoSysGain -autoSsys -av Average noise source parameters in time -f <file> Define input filename -scal <string> Specify Scal information file -scale <value> Scale the noise source parameters by value

Output	Interactive, graphical plot
Interactive key presses	1,2,3,4,5,6 and 7 Select plot type: 1. Noise source ON spectra for polarisation channels 1 and 2 2. Noise source ON and OFF spectra for polarisation channel 1 3. OFF/(ON-OFF) 4. Ssys (Jy) 5. Differential gain 6. Differential phase 7. ON and OFF spectra for polarisation channels 1 and 2 + - a u q w z Move to the next spectral dump Move to previous spectral dump Toggle averaging Unzoom - revert to original plot Quit Write the current spectrum to a text file Zoom-in on a region defined by the mouse
Example	<pre>sdhdf_plotNoiseCal -f uwl_210722_012146.hdf</pre> The initial plot is as follows, which shows the noise source ON spectra for polarisation 1 and 2:

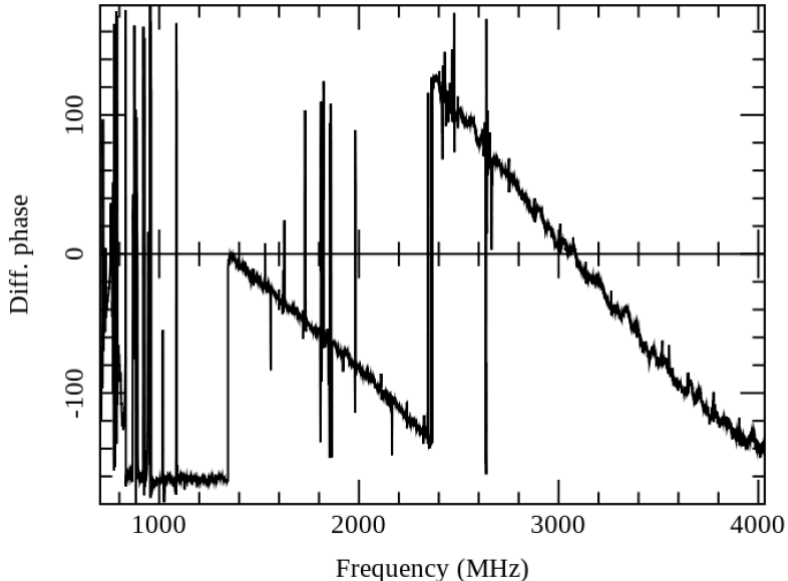


Pressing '3' (to select a plot of OFF/(ON-OFF)) and then 'z' to zoom in gives:



This is a plot of the system noise as a function of the noise source. If these values are multiplied by T_{cal} then this would provide T_{sys} .

More details on the noise source can be obtained using the other plot types. For instance, pressing '6' provides the differential phase of the noise source as shown below:

	spectral dump 0 
Known bugs	

sdhdf_calibrate

Purpose	Produces a flux density and/or polarisation calibrated data set for each input dat set	
File input	A list of input files is given on the command line.	
Command line arguments	-averageCal <f0> <f1> -e -h -nchanAstro <val> -norm -tcal	Use averaged noise source properties obtained between f0 and f1. Define the output file extension Help information Specify the number of astronomy channels to use when normalising the spectra Normalise the astronomy and noise source spectra Calibrate in K, instead of in Jy

	-tdumpAstro <val> Specify the astronomy spectral dump time (seconds)
Output	Output files (corresponding to each input file) that has been calibrated.
Interactive key presses	None
Example	sdhdf_calibrate *.hdf.T
Known bugs	

sdhdf_checkFile

Purpose	Allows the user to check for known issues with one of more data files. The checks are based on the observatory, receiver and date.
File input	Multiple files can be given on the command line.
Command line arguments	None
Output	Text that describes known issues with each input file.
Interactive key presses	None
Example	sdhdf_checkFile uwl_210722_012146.hdf The output is: <pre>Report for uwl_210722_012146.hdf ... sdhdf version: 2.0 ... searching history log for observatory: Parkes WARNING: The data tables were recorded with 5-dimensions. After MJD 59425 we record 4-dimensions (without beam number). WARNING: Topocentric frequency labelling in the SDHDF files incorrect (for more details see /u/hob044/software/new_c/gitRepos/sdhdf_tools/c/sdhdf Proc/runtime//observatory/parkes/historyLog/topoFreqE rror.txt)</pre>
Known bugs	
Runtime directory	This package relies on <pre>\$SDHDF_RUNTIME/observatory/<observatory>/historyLog/h istoryLog.dat</pre>

	This is a text file that contains a list of known issues at the specified observatory.
--	--

sdhdf_convertFromFITS

Purpose	Program to convert an SDFITS or PSRFITS file into SDHDF
File input	The -f command line option provides the input file
Command line arguments	-bin <val> Select specific phase bin to select from a PSRFITS file -cal Add noise source information into the SDHDF file -calOff <val1> <val2> Set noise source off values to VAL1 and VAL2 for two polarisations -calOn <val1> <val2> Set noise source on values to VAL1 and VAL2 for two polarisations -f Input FITS file -pulseOff <val1> <val2>. Define the pulse off region in PSRFITS file to VAL1 and VAL2 -o <filename> Set the output filename -type <val> Set the FITS type: 1 SDFITS 2 PSRFITS
Output	SDHDF file
Interactive key presses	None
Example	
Known bugs	

sdhdf_convertTo

Purpose	Program to convert SDHDF files into a different format (e.g., SDFITS)										
File input	Multiple input files can be listed on the command line										
Command line arguments	<table><tr><td>-1scanPerFile</td><td>Make separate files for each scan</td></tr><tr><td>-e <ext></td><td>Output file extension</td></tr><tr><td>-h</td><td>Help information</td></tr><tr><td>-noXpol</td><td>Do not write Xpol information</td></tr><tr><td>-srcChange <nameFrom> <nameTo></td><td>Change source names from "nameFrom" to "nameTo"</td></tr></table>	-1scanPerFile	Make separate files for each scan	-e <ext>	Output file extension	-h	Help information	-noXpol	Do not write Xpol information	-srcChange <nameFrom> <nameTo>	Change source names from "nameFrom" to "nameTo"
-1scanPerFile	Make separate files for each scan										
-e <ext>	Output file extension										
-h	Help information										
-noXpol	Do not write Xpol information										
-srcChange <nameFrom> <nameTo>	Change source names from "nameFrom" to "nameTo"										
Output	Output SDFITS files for each input file										
Interactive key presses	None										
Example											
Known bugs											

sdhdf_describe

Purpose	Provides information on the metadata within an sdhdf file. For instance, this is used to identify the source name, or project identifier for a set of data files.						
File input	Multiple sdhdf files can be listed on the command line. They can be placed in any position on the command line (i.e., before or after the other command line arguments)						
Command line arguments	<table><tr><td>-atoa</td><td>provides the meta-data in a format required for ingest into the ATOA data archive.</td></tr><tr><td>-attributes</td><td>Show table attributes</td></tr><tr><td>-band</td><td>shows meta-data relating to each sub-band for the astronomy data streams</td></tr></table>	-atoa	provides the meta-data in a format required for ingest into the ATOA data archive.	-attributes	Show table attributes	-band	shows meta-data relating to each sub-band for the astronomy data streams
-atoa	provides the meta-data in a format required for ingest into the ATOA data archive.						
-attributes	Show table attributes						
-band	shows meta-data relating to each sub-band for the astronomy data streams						

	-cal Provide information relating to the noise-source (e.g., duty-cycle and phase) -cband shows meta-data relating to each sub-band for the noise source -cdump Show meta-data relating to each spectral dump for the noise source -dump Show meta-data relating to each spectral dump for the astronomy data -h Provides help information relating to this package -history Shows information on how each file was processed -software Show information on the software used when producing each file -sb <N> Select only sub-band N
Output	The output is text written on the screen providing information about each of the input data files.
Interactive key presses	None
Example	sdhdf_describe -band uwl_210722_012146.hdf Gives output similar to <pre> ----- File Beam UTC SDHDF PID Sched_ID Source Tel Observer RCVR Bands POL_TYPE CAL M_Time RA_s DEC_s ----- uwl_210722_012146.hdf 0 2021-07-22-01:21:46 2.0 P737 29426 G300.969+1.147 Parkes mad134 UWL 26 AABBCRCI ON 99.9 12:34:53.00 -61:39:40.00 ----- # BandID F0 F1 NCHAN TDUMP NPOL NDUMP TOBS [Band] 000 band_SB0 704.00 832.00 262144 19.988 4 5 99.942 uwl_210722_012146.hdf [Band] 001 band_SB1 832.00 960.00 262144 19.988 4 5 99.942 uwl_210722_012146.hdf [Band] 002 band_SB2 960.00 1088.00 262144 19.988 4 5 99.942 uwl_210722_012146.hdf </pre>
Known bugs	

sdhdf_extractBand

Purpose	To extract sub-bands, or zoom-bands, out of sdhdf files to produce smaller files
File input	Input files listed on the command line

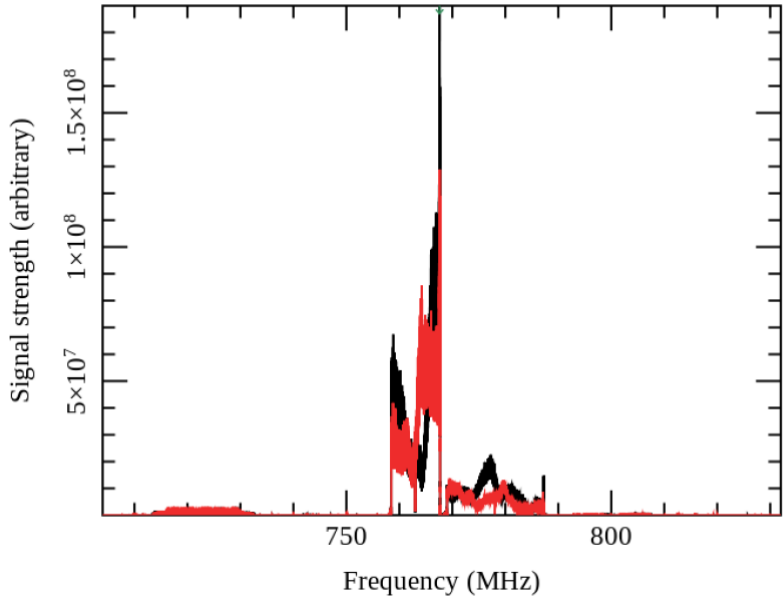
Command line arguments	-b <val> (or -band <val>, or -sb <val>) Specify which sub-band to extract (note that multiple -b options may be used to select multiple bands) -e <ext> Define the file extension of the output files -h Provide help information -zoom <f0> <f1> Produce a zoom-band between F0 and F1
Output	Output file for each input file only containing the specified bands
Interactive key presses	none
Example	<code>sdhdf_extractBand -b 5 -b 7 -b 13 -zoom 1664 1668 *.hdf</code>
Known bugs	

sdhdf_extractDump

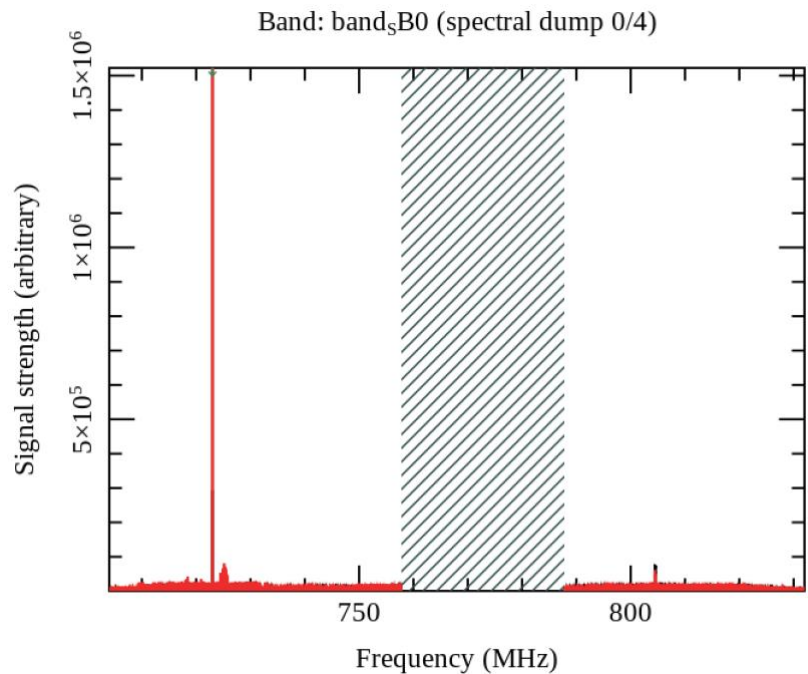
Purpose	Extract specific spectral dumps from the input files
File input	Input files listed on the command line
Command line arguments	-d <val> Specify the dump to extract (note multiple -d options can be used to extract more than one spectral dump) -e <ext> Define the extension of the output files
Output	Output file for each input file only containing the specified spectral dumps
Interactive key presses	None
Example	<code>sdhdf_extractDump -e dump -d 0 -d 1 *.hdf</code>
Known bugs	

sdhdf_flag

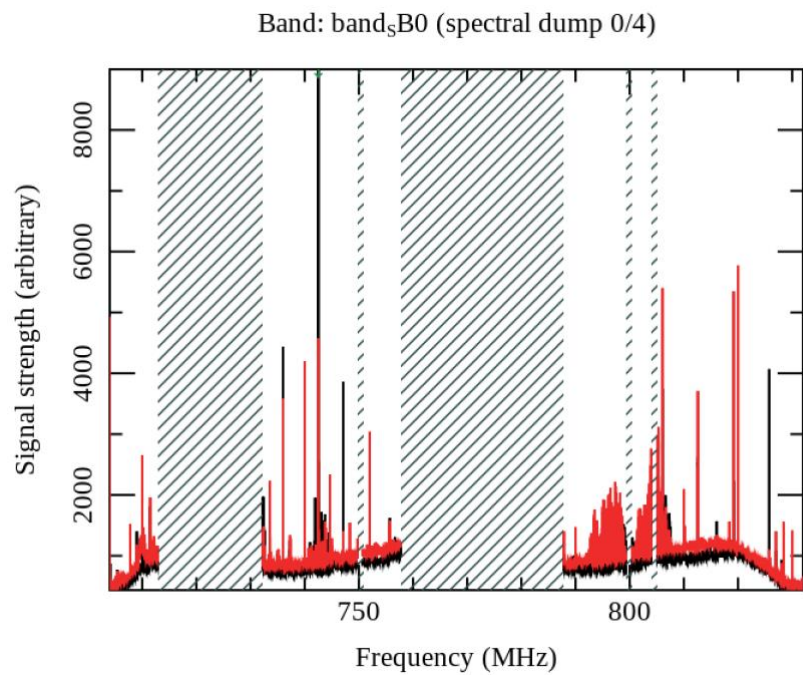
Purpose	Interactive flagging of data sets.
File input	Single input file defined using the -f <filename> command line option
Command line arguments	-f <filename> Define the input sdhdf file -h Provide help information -sb <val> (or -band <val>) Specify a specific sub-band for initial plotting -setnband <val> Set the maximum number of sub-bands for inclusion
Output	Output file that includes the weights set to zero for flagged data
Interactive key presses	1 Select polarisation channel 1 2 Select polarisation channel 2 + Move to the next spectral dump - Move to the previous spectral dump > Move to the next sub-band < Move to the previous sub-band a Toggle between zapping all spectral dumps versus only the currently selected one b Zap 5% of the sub-band edges d Toggle plotting the flagged regions f (also Z) Flag a region defined by the mouse clicks h Help information H Horizontal selection. This shows one polarisation and any points above the horizontal line will be flagged. This interface provides further options: Left mouse click: flag all spectral channels above the line 1,2,3,4,5 - set polynomial coefficients when removing baseline

	f - fit and remove a polynomial q - quit from this selection j Choose which spectral dump to move to k Flag all the channels in the current zoom region and move the to next spectral dump K Flag the entire spectral dump and move to the next one (all channels in all sub-bands) p Remove persistent RFI for this observatory q Quit r Remove the flagging of the data in region specified by mouse clicks s Save file t Automatically remove Parkes transient RFI u Un-zoom the plot z Specify a region to zoom-in to Z (also f) Flag a region defined by the mouse clicks
Example	<code>sdhdf_flag -f uwl_210722_012146.hdf</code> That bring up the following plot: Band: band_sB0 (spectral dump 0/4) 

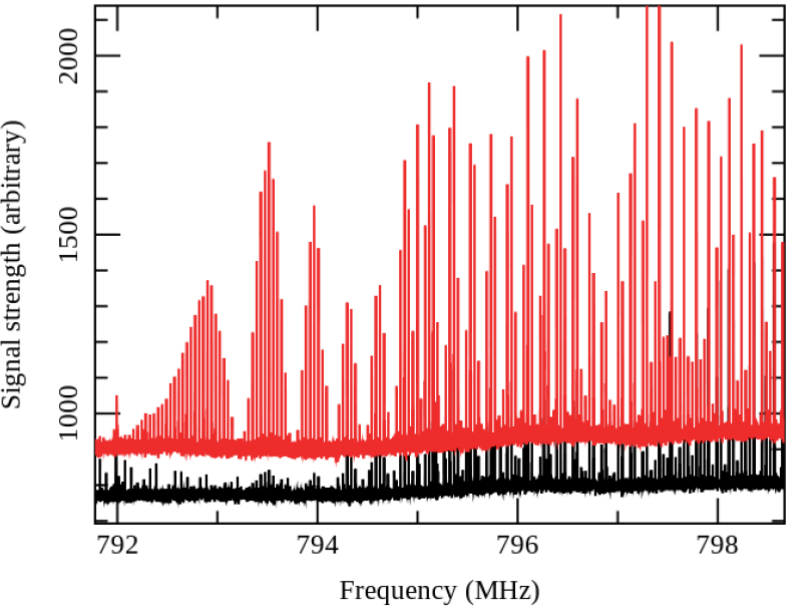
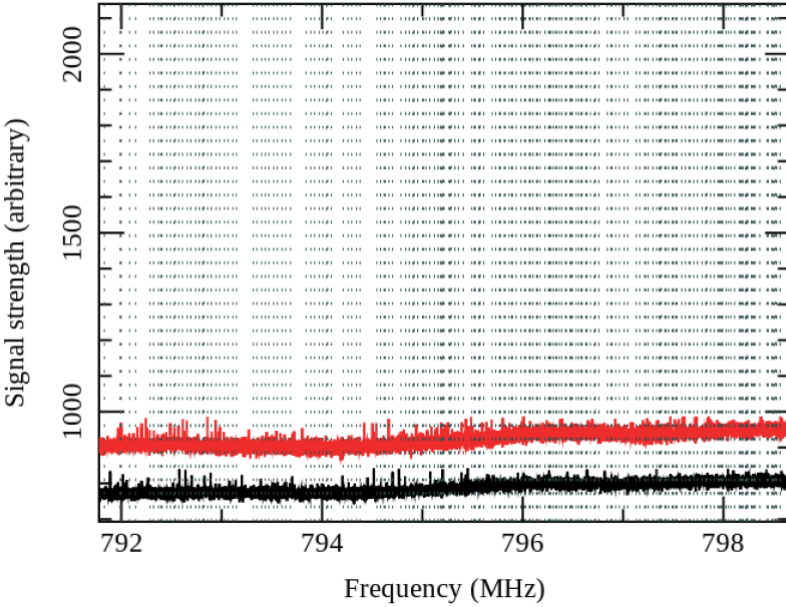
Moving the mouse cursor just to the left of the strong RFI and pressing Z allows a region to be selected and that RFI to be flagged



This process can be repeated to clean the band:



Note that a region (such as the region around 800 MHz) can be zoomed into using 'z':

	Band: band_sB0 (spectral dump 0/4)  It is clear that a comb of RFI exists. To clean the comb we can press H, move the horizontal line to remove the spikes, but not the baseline, click the mouse button and then press q to exit from that method of selection. The comb is now removed: Band: band_sB0 (spectral dump 0/4)  Press 's' to save the current flagging information and 'q' to quit.
Known bugs	

sdhdf_identify

Purpose	Identifies files based on specific properties (such as source name, observer, sky position etc.)	
File input	Multiple files can be listed on the command line.	
Command line arguments	-allBand Looks for matches amongst all sub-bands in the files (otherwise just picks the sub-band defined by -sb) -cal <id> Selects based on the noise source (usually ON or OFF) -coord <ra0> <dec0> <rad> Selects files within the specified angular radius of the position -h Provides help information -hdrversion <num> Selects files based on the SDHDF header version -info Displays more information on output lines -npol <num> Selects based on the number of polarisation channels (1, 2 or 4) -o <filename> Writes the output to the specified file -observer <name> Selects based on the observer name -pair Produce paired 'on' and 'off' observations - need -on and -off to be set -partial Does not require an exact match when checking source names -rcvr <name> Select based on receiver name -sb <num> Select sub-band number -sched <name> Selects based on schedule ID -src <name> Selects based on source name -tel <name> Selects based on telescope/observatory name -nband <num> Selects based on number of sub-	

	bands -ndump <num> Selects based on the number of spectral dumps -on <name> Defines the 'on' source -off <name> Defines the 'off' source
Output	Text to the screen or to the output file giving the files that are matched to the input arguments
Interactive key presses	None
Example	<code>sdhdf_identify -o pairSrcRef.dat -partial -pair -on S -off R uwl*.selectBands.T</code>
Known bugs	

sdhdf_join

Purpose	Package to join two sdhdf files together (currently in the time dimension)
File input	List of input files given on the command line
Command line arguments	-buildFreq Build up a file which includes multiple sub-bands through multiple input files -dsb <num> <str> Define the sub-band "num" to be the spectrum in file with name "str" -o <filename> Output file name
Output	Output file (as specified by -o) that is all the input files joined together
Interactive key presses	None
Example	<code>sdhdf_join -o join.hdf *.hdf.T</code>
Known bugs	

sdhdf_listLines

Purpose	Program that lists the reference frequencies of known spectral lines
File input	None
Command line arguments	If no command line arguments are present then the code lists spectral lines between 704 and 4032 MHz (specified from the Parkes UWL receiver band). Other the command line arguments provide the frequency range to search
Output	Text on the terminal listing the known spectral lines. The spectral line rest frequencies are obtained from \$SDHDF_RUNTIME/spectralLines/spectralLines.dat
Interactive key presses	None
Example	<pre>\$ sdhdf_listLines 1600 1700 1610.2469 CH_3OCHO 1610.9000 CH_3OCHO 1612.2310 OH: Hydroxyl radical 1624.5179 ^17OH 1626.1610 ^17OH 1637.5640 ^18OH 1638.8051 HCOOH 1639.5031 ^18OH 1665.4019 OH: Hydroxyl radical 1667.3590 OH: Hydroxyl radical 1692.7950 ^18OH</pre>
Known bugs	

sdhdf_modify

Purpose	Modify one or more sdhdf files to produce new, processed files. For instance, this routine can frequency average data, average in time, convert to the local standard of rest etc.
File input	List of input files given on the command line
Command line arguments	-bary Convert to barycentric reference frame -calBin <val> Number of phase bins when the noise source was recorded

-calFitN <val>	
-cc_eps1 <val>	
-cc_eps2 <val>	
-cc_phi1 <val>	
-cc_phi2 <val>	
-divideAstro	
-divideCal	
-e <ext>	Define the output file extensions, which get appended to the input file names
-ephem <string>	Define solar system ephemeris to use for barycentric/LSR corrections
-F	Average all data in frequency
-fav <val>	Frequency average “val” channels
-fchbw <val>	Frequency average to specified channel bandwidth
-id <val>	Ignore specified spectral dump (multiple entries can be provided)
-interp <val>	
-lsr	Convert to the local standard of rest reference frame
-muellerI	
-nch <val>	Frequency average to specific number of channels per sub-band
-origNchan <val>	Specify the original number of channels when the observation was carried out
-p1	Produce $AA^* + BB^*$
-p2	
-p3	
-para <val>	Define the parallactic angle of the observation (degrees)
-psum	

	ON) -batch <filename> Provides a list of ON and OFF file names in a text file. -h Help information -off <filename> Filename defining the off source file -on <filename> Filename defining the on source file -scaleOff <string> Filename to text file providing scaling factors -sclA <val> Value to scale polarisation A -sclB <val> Value to scale polarisation B -subtract Produce ON-OFF. The default is (ON-OFF)/OFF
Output	One output file (defined using -o) or multiple output files (from the -batch option)
Interactive key presses	None
Example	
Known bugs	

sdhdf_map

Purpose	
File input	
Command line arguments	
Output	
Interactive key presses	
Example	
Known bugs	

sdhdf_pointing

Purpose	
File input	
Command line arguments	
Output	
Interactive key presses	
Example	
Known bugs	

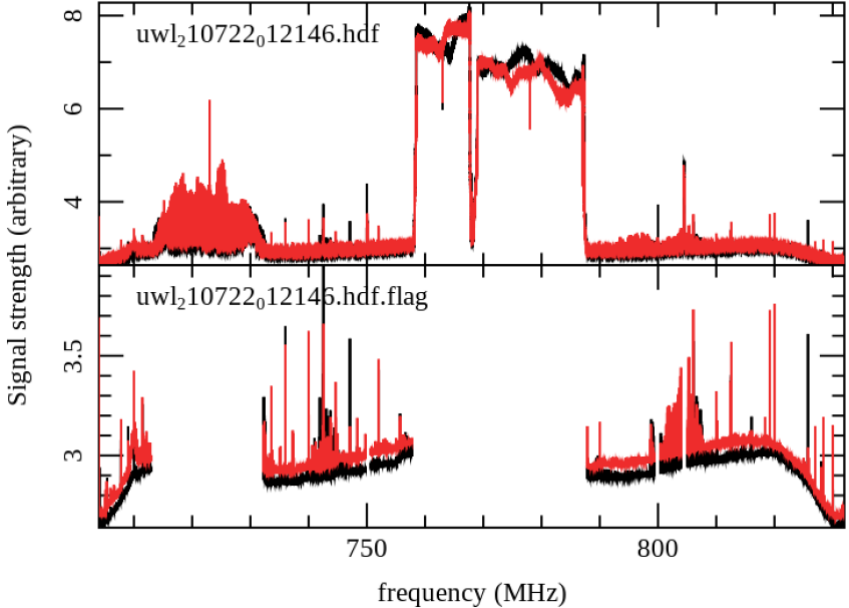
sdhdf_plotMap

Purpose	
File input	
Command line arguments	
Output	
Interactive key presses	
Example	
Known bugs	

sdhdf_plotMultiSpec

Purpose	Software to plot multiple spectra (e.g., different transitions of a spectral line, or ON and OFF source pointings)
File input	Each file is listed on the command line. Note that the ordering is important as the properties of the plot for each file must come prior to the file name.
Command line arguments	-4pol Plot 4 polarisations (default is 2) -av <val> Frequency average “val” channels before plotting

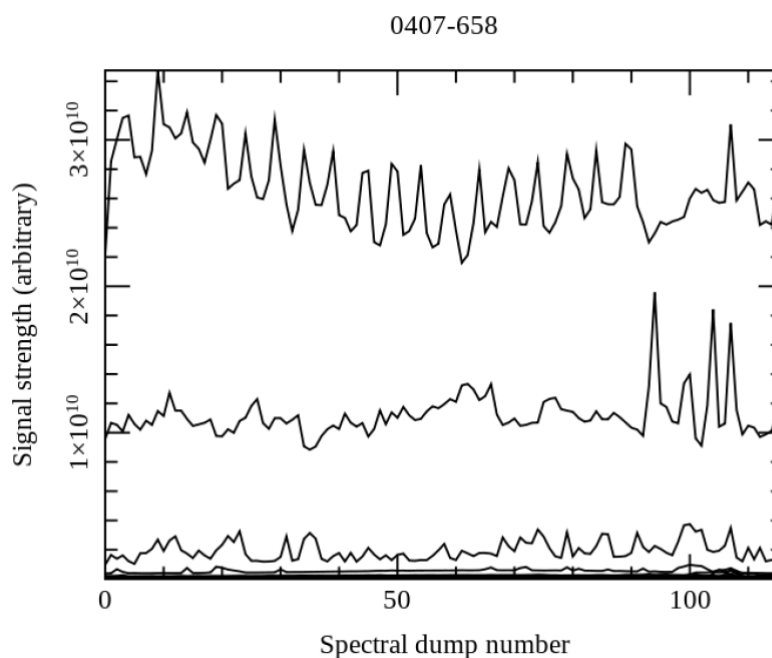
	-bl <f0> <f1> Remove a baseline (simple mean value) obtained between frequencies F0 and F1 -h Provide help information -f0 <f0> Start from F0 (MHz) when plotting -f1 <f1> Finish the plot at F1 (MHz) -flipV Flip the sign of Stokes V (polarisation 4) -fref <val> Define the rest frequency. If set then the x-axis will be plotted in velocity instead of frequency. -g <text> Define the PGPLOT/GIZA graphical device -hline <val> Draw a horizontal line at y-coordinate "val" -join Join the plots together (by default they are shown on different plots) -locky <y1> <y2> Lock the y-axis for all plots to the specified values -log Plot on a logarithmic y-axis -nx <val> Split the figure into "val" panels in the x-direction -ny <val> Split the figure into "val" panels in the y-direction -p Average in polarisation -query <filename> A query will be made after each spectrum and recorded into "filename" -RLCP Plot right and left-hand circular polarisation (assumes calibrated data) -sb <val> Choose specific sub-band -sd <val> Choose specific spectral dump -vline <val> Draw a vertical line with x-coordinate "val" -ylabel <text> Change the y-axis label to "text" -yscale <val> Scale the y-coordinates by "val"
Output	Figure showing one or more spectra

Interactive key presses	None
Example	<pre>sdhdf_plotMultiSpec -log -join -ny 2 uwl_210722_012146.hdf uwl_210722_012146.hdf.flag</pre> The above command will make the following plot where the compare pre- and post-flagging data files 
Known bugs	

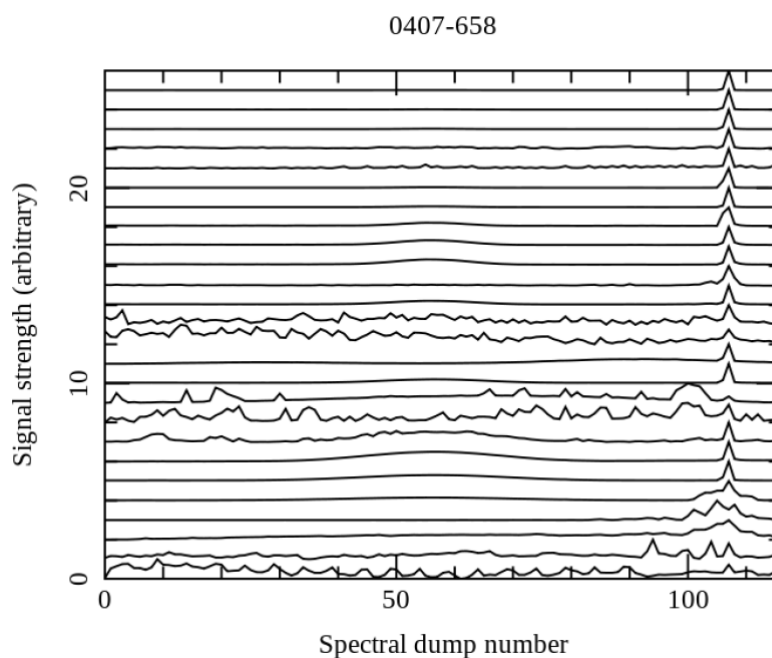
sdhdf_plotScan

Purpose	Provides the method to plot spectra recorded in scanning mode (i.e., expected variations as a function of time)
File input	Single input file specified using the -f option
Command line arguments	-divWeights Divide spectra by weight information -f <name> Specify the input file name -freqRange <f0> <f1> Analyse the specified (in MHz) frequency range -src <name> Specify the source name (important for scans across known flux)

	calibrators)
Output	Interactive plot
Interactive key presses	+ Select next sub-band - Select previous sub-band f Provide information per sub-band of the model fit l Toggle plotting on a logarithmic scale L Toggle plotting labels n Normalise each scan for each sub-band o Offset each scan for each sub-band from each other p Specify zero coordinate q Quit s Select current sub-band (defined using + and -) u Unzoom to original plot x Change the x-axis (keep pressing x to change between different axes) • Spectral dump number • Time elapsed from start • Right ascension (degrees) • Declination (degrees) • Azimuth (degrees) • Elevation (degrees) • Angle from coordinate zero (specified using 'p' or from the source name) • Right ascension from coordinate zero • Declination from coordinate zero z Zoom in on region defined using the mouse
Example	<pre>sdhdf_plotScan -src 0407-658 -f uwl_200720_024617.hdf</pre> This will plot the results of a scan across 0407-658. Note that the source name in the file is not the name of the primary flux calibrator and therefore this is specified on the command line using -src. The initial output plot is as follows and note that no RFI mitigation has been applied.

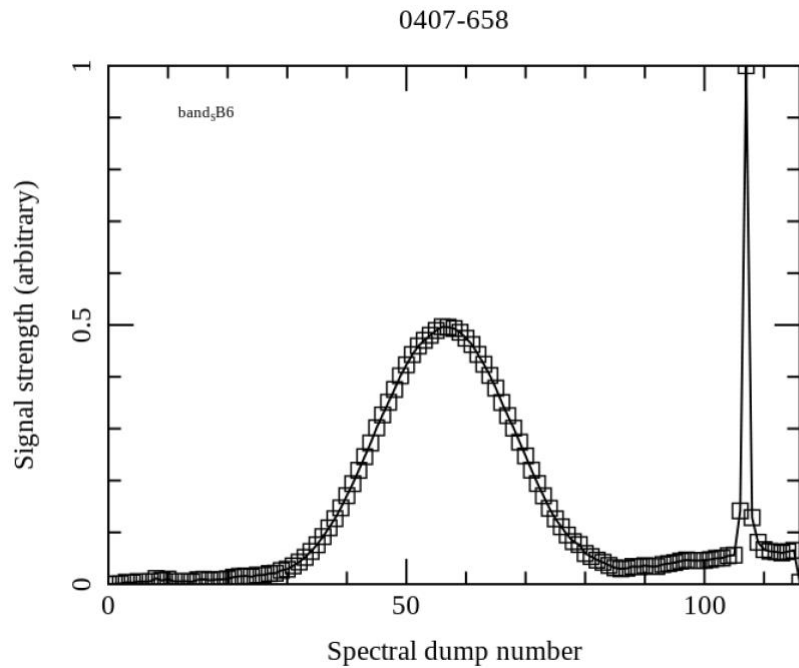


Each trace shows the continuum flux (arbitrary units) as a function of spectral dump number. Note that this is dominated by the RFI in the lowest bands. We can make the sub-bands clearer by normalising (pressing 'n') and then offsetting them (pressing 'o').

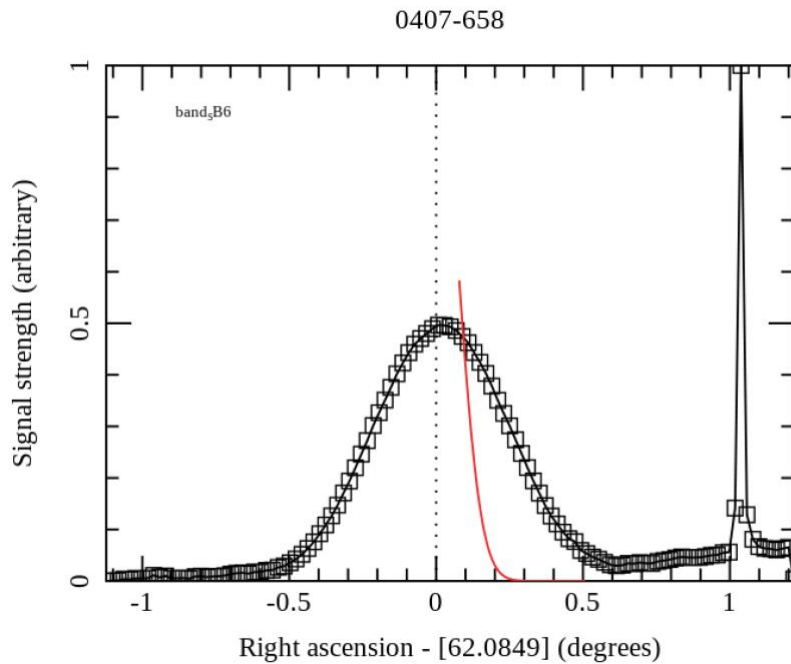


We note some strong, broadband RFI near the end (possibly from saturation). Note that if we used `sdhdf_flag` or `sdhdf_autoFlag` to remove the RFI then these figures would be improved. For now

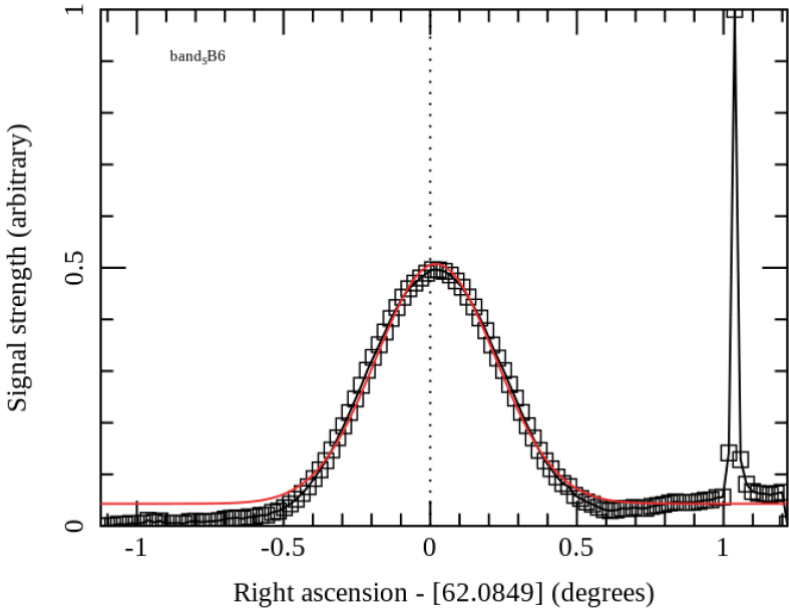
let's select sub-band 7 by pressing the '+' symbol seven times and then press 's':



Now the primary source is clear (and we could remove the RFI to improve the figure further). If we press 'x' seven times then we can plot the offset from the known right ascension of 0407-658:



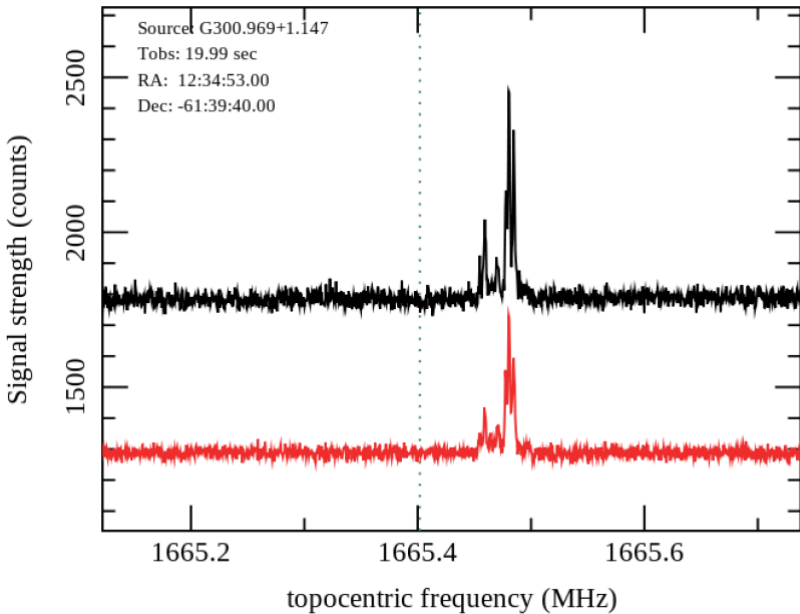
And press 'f' to fit a Gaussian curve:

	0407-658  The output (on the terminal) provides the properties of the Gaussian fit for each sub-band.
Known bugs	

sdhdf_plotSpectrum

Purpose	To plot and interact with spectra	
File input	This program only plots spectra from a single file, which is defined using the -f command line argument.	
Command line arguments	-f <file>	File name of sdhdf file to plot
	-fref <num>	Define a reference frequency for plotting the spectra with a velocity axis
	-h	Provide help information
	-g <type>	PGPLOT graphical device definition
	-sb <num>	Define the spectral band to plot
	-sd <num>	Define the spectral dump to plot

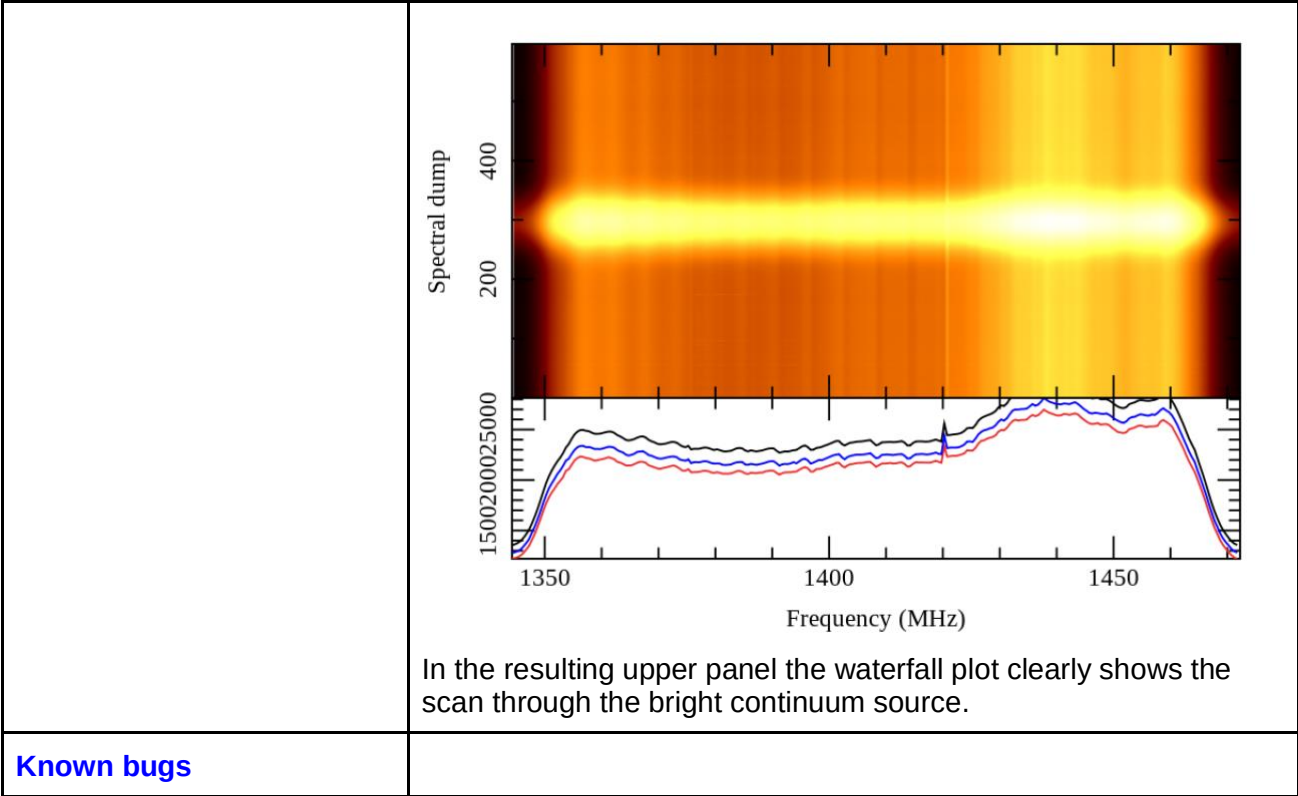
	-yunit <text> Define the unit that will be written on the y-axis label.																																						
Output	The code produces an interactive plot and various text output on the terminal.																																						
Interactive key presses	The following key strokes can be used in the interactive window: <table> <tr> <td>Left mouse click (or A)</td><td>Identify the coordinate for the mouse click</td></tr> <tr> <td>></td><td>Move to next sub-band</td></tr> <tr> <td><</td><td>Move to previous sub-band</td></tr> <tr> <td>+</td><td>Move to next spectral dump</td></tr> <tr> <td>-</td><td>Move to previous spectral dump</td></tr> <tr> <td>f</td><td>Toggle flagging</td></tr> <tr> <td>h</td><td>help</td></tr> <tr> <td>l</td><td>Toggle plotting on a logarithmic or linear scale</td></tr> <tr> <td>L</td><td>List the data on the screen corresponding the polarisation 1.</td></tr> <tr> <td>m</td><td>Toggle plotting rest frequencies of molecular lines</td></tr> <tr> <td>o</td><td>Output the spectra in the current zoom-region to a text file</td></tr> <tr> <td>p</td><td>Toggle plotting 2 polarisations (AA and BB) or 4 (all states)</td></tr> <tr> <td>q</td><td>Quit from the code</td></tr> <tr> <td>r</td><td>Toggle plotting rest frequencies of recombination lines.</td></tr> <tr> <td>s</td><td>Provide statistics from a selected region.</td></tr> <tr> <td>u</td><td>unzoom (remove any zoom selection)</td></tr> <tr> <td>w</td><td>Toggle dividing (or not) by the weights</td></tr> <tr> <td>x</td><td>Toggle how the x-axis is displayed</td></tr> <tr> <td>z</td><td>Zoom-in on a particular part of the spectrum</td></tr> </table>	Left mouse click (or A)	Identify the coordinate for the mouse click	>	Move to next sub-band	<	Move to previous sub-band	+	Move to next spectral dump	-	Move to previous spectral dump	f	Toggle flagging	h	help	l	Toggle plotting on a logarithmic or linear scale	L	List the data on the screen corresponding the polarisation 1.	m	Toggle plotting rest frequencies of molecular lines	o	Output the spectra in the current zoom-region to a text file	p	Toggle plotting 2 polarisations (AA and BB) or 4 (all states)	q	Quit from the code	r	Toggle plotting rest frequencies of recombination lines.	s	Provide statistics from a selected region.	u	unzoom (remove any zoom selection)	w	Toggle dividing (or not) by the weights	x	Toggle how the x-axis is displayed	z	Zoom-in on a particular part of the spectrum
Left mouse click (or A)	Identify the coordinate for the mouse click																																						
>	Move to next sub-band																																						
<	Move to previous sub-band																																						
+	Move to next spectral dump																																						
-	Move to previous spectral dump																																						
f	Toggle flagging																																						
h	help																																						
l	Toggle plotting on a logarithmic or linear scale																																						
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Example	<pre>sdhdf_plotSpectrum -sb 7 -f uwl_210722_012146.hdf</pre> This shows the spectra (for polarisations AA and BB) in sub-band 7. We then zoomed-in (using the 'z' option) around the OH line 1665.4 MHz where the rest frequency of the line is plotted as a green, vertical line (by pressing 'm'). 
Known bugs	

sdhdf_waterfall

Purpose	Plot the spectra as a function of time (as a waterfall plot)
File input	Multiple files can be listed on the command line. These are concatenated when making the waterfall plot.
Command line arguments	-band <num> Specify sub-band number -crange <f0> <f1> Specify a frequency range which is “clean” and so can be used for baseline removal (in MHz) -frange <f0> <f1> Specify the frequency range for the plot (in MHz) -log Plot with a logarithmic signal strength axis

	-pol <num> Specify polarisation channel to plot -ndump <val> Set maximum number of spectral dumps to plot
Output	Interactive plot
Interactive key presses	Left mouse click (or A) Identify coordinate of mouse click and plot corresponding spectrum a Toggle selecting a spectrum c Toggle plotting colour and grey-scale images f Specify a new frequency range for the plot l Toggle plotting on a logarithmic scale q Quit r Set the range for the signal intensity (z-direction) s Remove a baseline defined by a specified frequency band u Unzoom to the original plot x Zoom-in on the x-direction y Zoom-in on the y-direction
Example	<pre>sdhdf_waterfall -band 5 uwl_210722_201153.hdf.f128</pre> This will make a waterfall plot for sub-band 5 in this file that is a scan across a bright continuum source.



sdhdf_plotWide

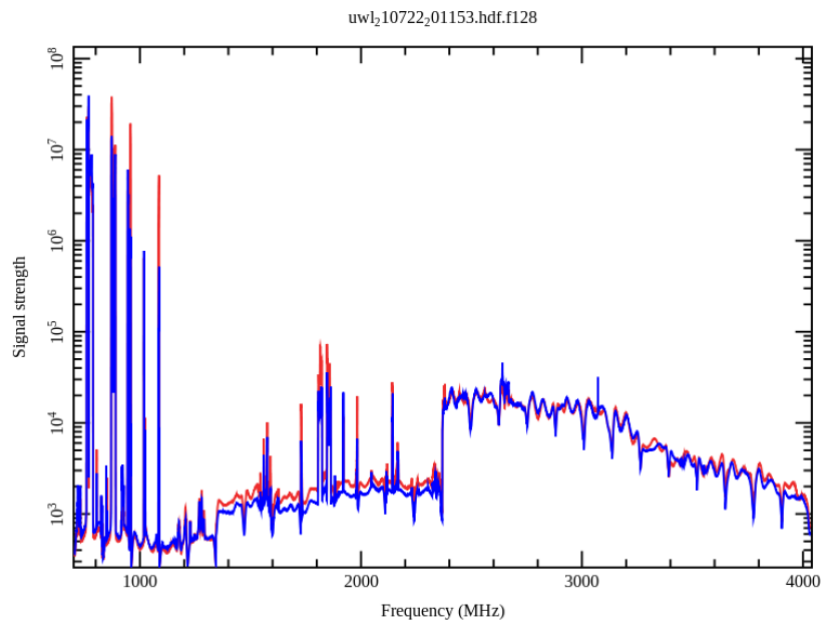
Purpose	Routine to make wide-bandwidth plots of the spectra.	
File input	Multiple files can be listed on the command line.	
Command line arguments	-4pol Plot 4 polarisation channels (default = 2) -ch <val> Set the character height for the labels on the plot -dispAz Display as a function of Azimuthal angle -flag <f0> <f1> Flag data between F0 and F1 (MHz) - this argument can be provided multiple times -g <string> Set the PGPLOT/GIZA graphics device -h Help information -maxhold Plot a max-hold spectrum	

	-maxx <val> Set the maximum frequency to plot (MHz) -minx <val> Set the minimum frequency to plot (MHz) -nolog Do not plot on a logarithmic scale -oCoeff <val> -sb <val> Specify the spectral band number -sd <val> Specify the spectral dump number -shade <f0> <f1> Shade (for highlighting purposes) the spectrum between F0 and F1 (this command line argument can be listed multiple times) -splitRF Split the wide-band into multiple panels -stokes Plot Stokes parameters -transmitters Label known RFI transmitters -waterfall Plot as a waterfall plot -ylabel <string> Define the y-axis label
Output	The output is an interactive plot
Interactive key presses	Left mouse click (or A) Identify current position of the mouse cursor h Help information m Toggle plotting the max-hold spectrum M Toggle plotting known rest frequencies of molecular lines q quit t Toggle plotting known RFI transmitters u Unzoom - revert to original plot w Define new ranges for the waterfall plot z Zoom in on region defined using the mouse cursor

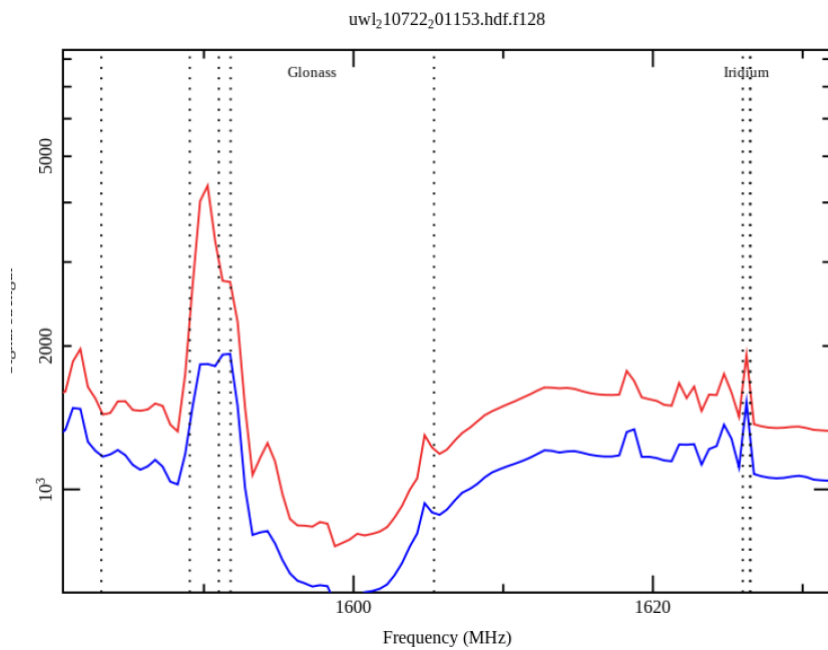
Example

```
sdhdf_plotWide uwl_210722_201153.hdf.f128
```

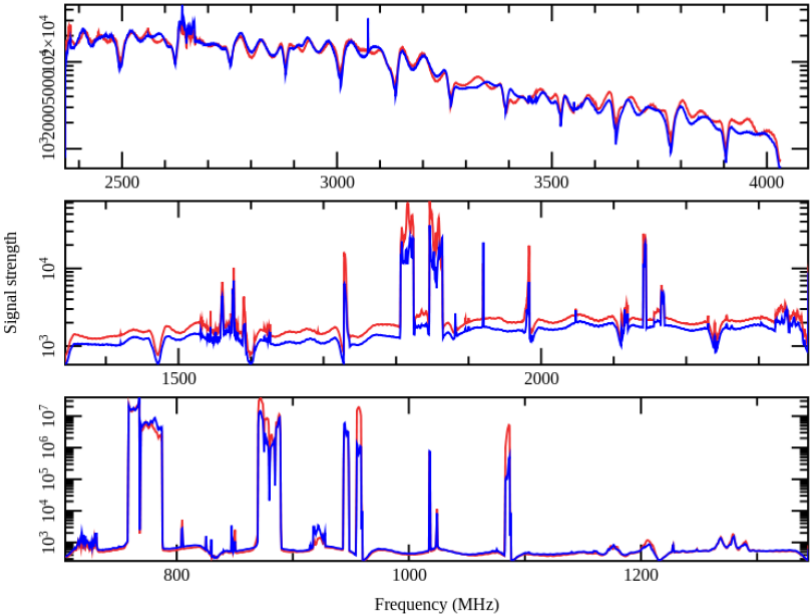
Provides a simple wide-band spectrum:



Pressing 't' to plot known RFI signals and then zooming-in using 'z' produces plots like:



The Parkes UWL 3-RF bands can be divided using:

	<pre>sdhdf_plotWide -splitRF uwl_210722_201153.hdf.f128</pre> 
Known bugs	

sdhdf_quickdump

Purpose	Provides text output of spectra - for use in other packages (such as python or gnuplot scripts)
File input	Multiple files given on the command line
Command line arguments	-bandRange <val1> <val2> Specify specific sub-band range. -cal Output the noise ON and OFF noise-source information -cal32 Output the 32-binned noise source information -cal32_ch <val> Select specific channel for the noise source for the 32 bin data

	-cal32_tav Average the 32 bin data in time -dumpRange <val1> <val2> Specify dump range for output -e <file extension> Provide output with specified file extension -freqRange <f0> <f1> Specify frequency range (in MHz) for output -h Help information -fref <val> Provide rest frequency value for determining velocities -tsys
Output	Text output to the terminal or to an output file.
Interactive key presses	None
Example	sdhdf_quickdump uwl_210722_012146.hdf This outputs text similar to: <pre>uwl_210722_012146.hdf 0 0 0 22227 714.853088 1000.68 1084.39 -12.5368 -61.7926 9760 uwl_210722_012146.hdf 0 0 0 22228 714.853577 2160.8 1359.15 -36.8301 -350.773 9760 uwl_210722_012146.hdf 0 0 0 22229 714.854065 1382.23 1215.18 -33.8768 -221.356 9760 uwl_210722_012146.hdf 0 0 0 22230 714.854553 1021.48 1083.72 -16.1586 -51.1686 9760</pre> The default output columns are: 1. Filename 2. Beam number 3. Sub-band number 4. Spectral-dump number 5. Channel number 6. Frequency 7. Pol 1 8. Pol 2 9. Pol 3 10. Pol 4 11. Channel weight

	This could be output to a text file on disk, or processed, using standard linux commands: For instance to extract the frequency and first two polarisations can use: <pre>sdhdf_quickdump uwl_210722_012146.hdf awk '{print \$6,\$7,\$8}' > myfile.dat</pre>
Known bugs	

sdhdf_sum

Purpose	Routines to sum spectra together
File input	All input files listed on the command line
Command line arguments	-divN Divide the output spectra by the number of input files (i.e., forms a mean, instead of a sum) -h Provide help information -o <filename> Provide the output name
Output	Single output file (as defined using the -o command line option)
Interactive key presses	None
Example	<code>sdhdf_sum -o sum.hdf uwl*.hdf</code>
Known bugs	

sdhdf_statistics

Purpose	Provide basic statistical information from input spectra
File input	Multiple filenames
Command line arguments	-h Help information

	-freqRange <f0> <f1> Specify frequency range (in MHz) -sb <val> Specify spectral dump number
Output	Text on the terminal presenting statistical information. The output columns are: 1. Label 2. Filename 3. Source name 4. Spectral dump number 5. Mean value for pol 1 6. Mean value for pol 2 7. Minimum value for pol 1 8. Maximum value for pol 1 9. Minimum value for pol 2 10. Maximum value for pol 2 11. MJD of centre of spectral dump
Interactive key presses	None
Example	<pre>sdhdf_statistics -sb 5 -freqRange 1420 1430 uw1_210722_012146.hdf</pre>
Known bugs	